the rotation by 180^0 around an axis that is perpendicular to two opposite edges, (12)(34) is the rotation by 180^0 around an axis that is perpendicular to two opposite faces, (123) is the rotation around a main diagonal by 120^0, and (1234) is the rotation by 90^0 around an axis that is perpendicular to two opposite faces; this allows us to compute the traces easily, using the fact that the trace of a rotation by the angle ϕ in $\mathbb{R}^3$ is $1 + 2\cos\phi$. Now the character of $\mathbb{C}^3_-$ is found by multiplying the character of $\mathbb{C}^3_+$ by the character of the sign representation.

Finally, we explain how to obtain the character table of A_5 (even permutations of five items). The group A_5 is the group of rotations of the regular icosahedron. Thus it has a 3-dimensional "rotation representation" $\mathbb{C}^3_+$, in which (12)(34) is the rotation by 180^0 around an axis perpendicular to two opposite edges, (123) is the rotation by 120^0 around an axis perpendicular to two opposite faces, and (12345), (13254) are the rotations by 72^0, respectively 144^0, around axes going through two opposite vertices. The character of this representation is computed from this description in a straightforward way.

Another representation of A_5, which is also 3-dimensional, is $\mathbb{C}^3_+$ twisted by the automorphism of A_5 given by conjugation by (12) inside S_5. This representation is denoted by $\mathbb{C}^3_-$. It has the same character as $\mathbb{C}^3_+$, except that the conjugacy classes (12345) and (13245) are interchanged.

There are two remaining irreducible representations, and by the sum of squares formula their dimensions are 4 and 5. So we call them $\mathbb{C}^4$ and $\mathbb{C}^5$.

The representation $\mathbb{C}^4$ is realized on the space of functions on the set $\{1, 2, 3, 4, 5\}$ with zero sum of values, where A_5 acts by permutations (check that it is irreducible!). The character of this representation is equal to the character of the 5-dimensional permutation representation minus the character of the 1-dimensional trivial representation (constant functions). The former at an element g is equal to the number of items among 1, 2, 3, 4, 5 which are fixed by g.

The representation $\mathbb{C}^5$ is realized on the space of functions on pairs of opposite vertices of the icosahedron which has zero sum of